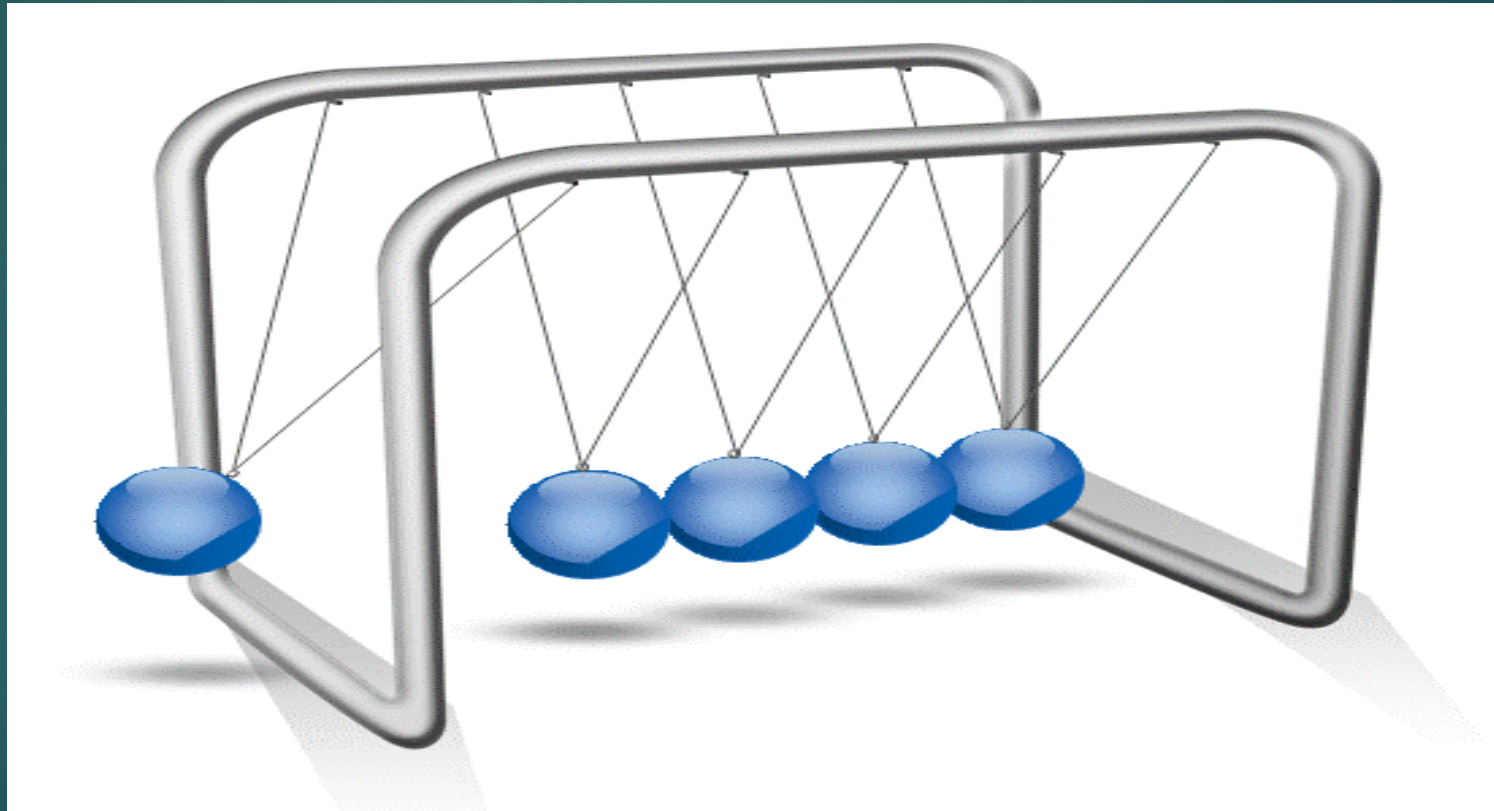


Experiment (4)

Conservation of Energy



Introduction

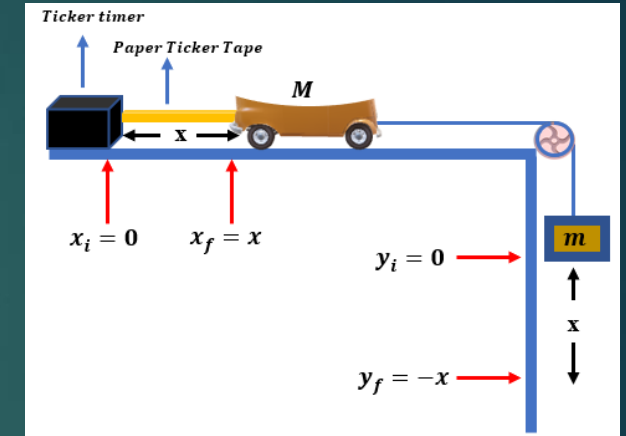
If all forces acting on a system are conservative forces, then the total mechanical energy is conserved.

$$E_i = E_f$$

Here, we investigate the validity of the principal of conservation of energy in an experiment consisting of two parts. During the first part, we will use a cart-mass system while in the second part, we will use a spring-mass system.

Theory Part 1:

The Ticker timer measures the time needed by the cart to travel a distance x . If the system is released from rest, then the mass m moves downward while dragging the cart M to the right. The distance traveled by the mass m is equal in magnitude to the distance traveled by the cart M . If we choose the initial positions of the cart M and the mass m as $x_i = 0$ and $y_i = 0$, then the final positions are $x_f = x$ and $y_f = -x$ (the positive direction along the the y-axis is upward).



$$E_i = E_f \dots\dots\dots (1)$$

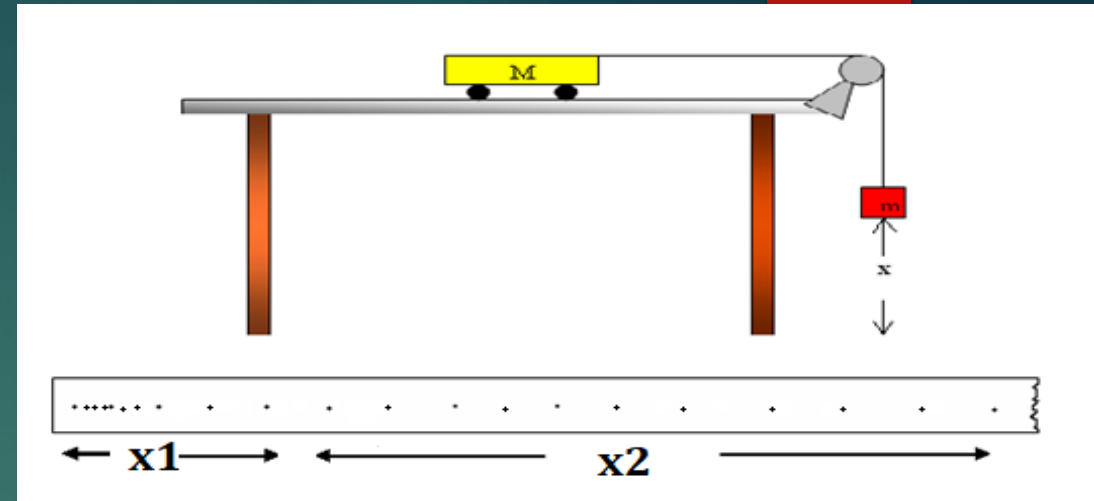
$$mgx = \frac{1}{2}Mv^2 + \frac{1}{2}mv^2 \Rightarrow x = \frac{M+m}{mg}v^2$$

$$v = \frac{2x}{t}$$

$$x = \frac{1}{2} \frac{mg}{M+m} t^2 \dots\dots\dots (2)$$

Procedure Part 1:

- Connect the mass m and the cart M by a string and let the string pass over the pulley.
 - Cut a piece of paper ticker tape about the length of the height of the cart from the floor and tape one end to cart.
 - Release m from rest and let it move a distance x .
 - Remove the paper tape from the cart.
 - Use the dots on the paper tape to measure the time and the distance traveled by m as shown below.
 - Tabulate your data in the table shown below.
-
- Plot x versus t^2 on a graph paper and draw the best straight line through the points.
 - Find the slope. Using equation 2 and the value of the slope, calculate the earth's gravitational acceleration.
 - Calculate the absolute and relative errors in the earth's gravitational acceleration.
 - What are the possible sources of errors in this part?



Note that the frequency of the ticker timer is 50 Hz which means that the time between every two points equals 0.02 seconds. Choose $x_0 = 0$ as a reference.

$m = (\quad)$		$M = (\quad)$	
Trial	$x(\quad)$	$t(\quad)$	$t^2(\quad)$
1			
2			
3			
4			
5			

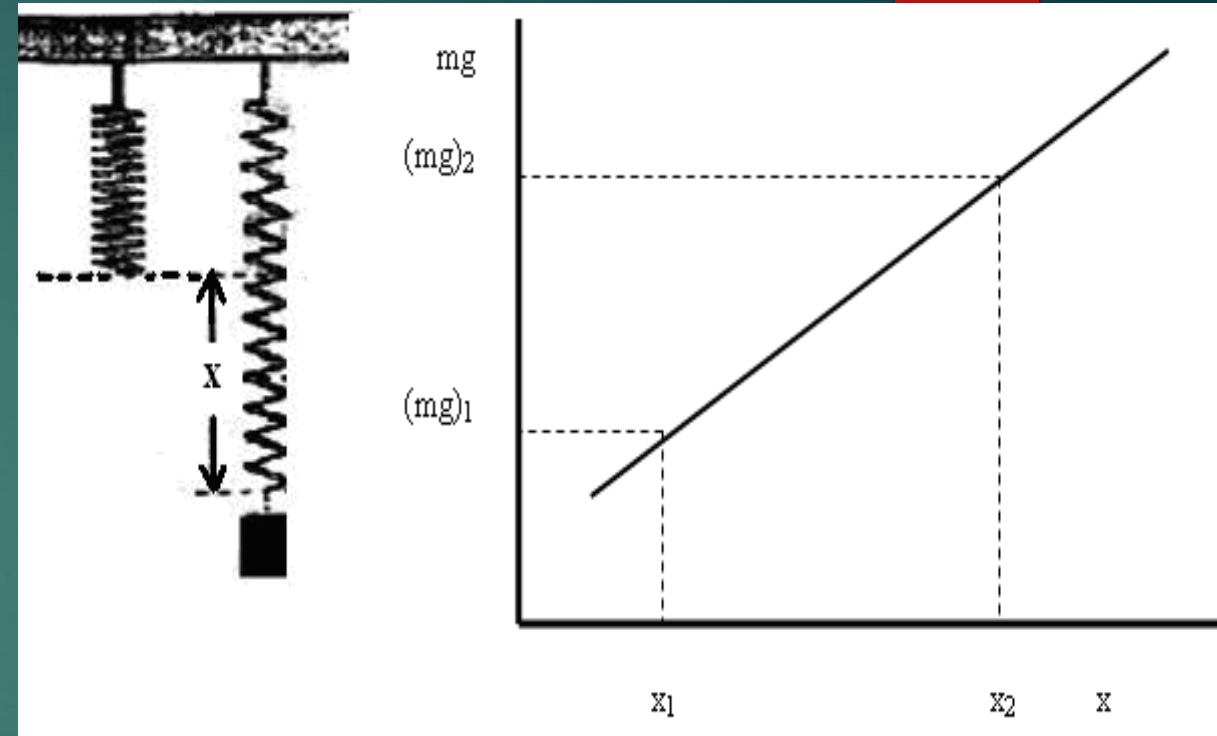
Theory Part 2:

To find the spring constant K

$$F = Kx$$

$$mg = Kx$$

$$\text{slope} = \frac{(mg)_2 - (mg)_1}{x_2 - x_1} = K \dots\dots\dots(3)$$



In conservation of energy the mass is dropped from the natural length and the spring extend to $X_{\max}$

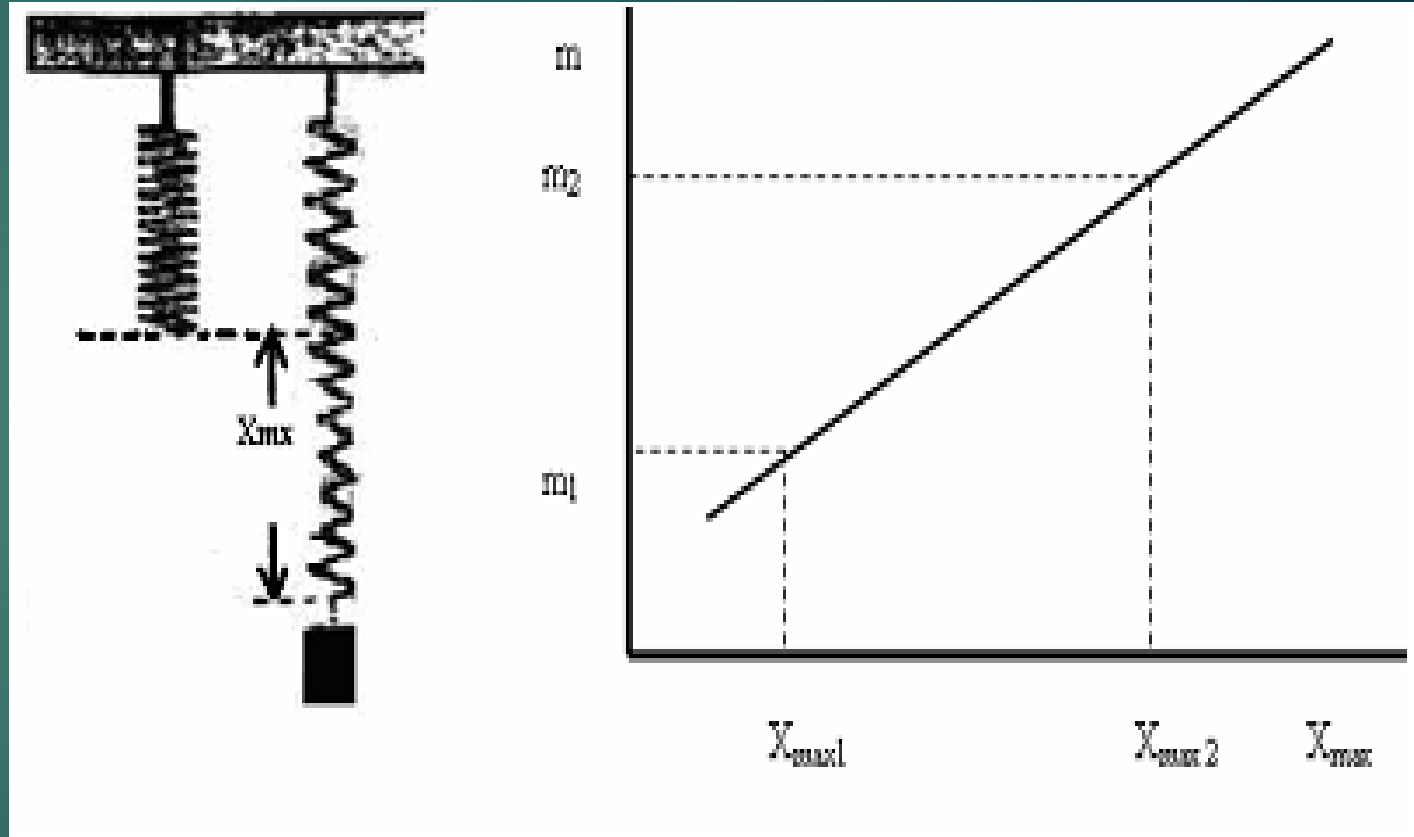
$$E_i = E_f$$

$$mgX_{\max} = \frac{1}{2} KX_{\max}^2$$

$$m = \left(\frac{K}{2g} \right) X_{\max}$$

$$\text{slope} = \frac{K}{2g}$$

$$g = \frac{K}{2 \times \text{slope}} \quad \text{.....(4)}$$



Procedure Part 2:

Calculating the force constant (K).

- Attach the mass m to the spring.
- Gradually release the mass m and wait till the mass comes to rest.
- Measure the equilibrium distance $x_{equilibrium}$.
- Change the mass and repeat the previous steps for five trials.
- Tabulate your results in the table shown below.
- Plot the mass m versus x on a graph paper and draw the best straight line through the points.
- Find the slope. Use **equation 3** and the calculated value of the **slope** to calculate the force constant k .
- Calculate the absolute and relative errors in the force constant k .

Verifying the concept of conservation of the mechanical energy.

- Attach the mass m to the spring.
- Release the mass to fall from its natural length and record its **maximum distance** from equilibrium x_{max} .
- Change the mass and repeat the previous steps for five trials.
- Tabulate your results in the table shown below.
- Plot the mass m versus x_{max} on a graph paper and draw the best straight line through the points.
- Find the slope. Use **equation 4** and the calculated value of the **slope** to calculate the earth's gravitational acceleration.
- Calculate the absolute and relative errors in the earth's gravitational acceleration.

<i>Trial</i>	$m(\quad)$	$x_{equilibrium}(\quad)$
1		
2		
3		
4		
5		

<i>Trial</i>	$m(\quad)$	$x_{max}(\quad)$
1		
2		
3		
4		
5		

THANK YOU
